MOLLI: Model Optimization via Likelihood-based Introgression

A continual learning framework for LLMs inspired by genetic introgression in Poecilia formosa (Schartl et al., Nature 2026)

Figure 1. The MOLLI evolution cycle, repeated for each new domain.

1. The Problem

When a language model is fine-tuned sequentially on multiple domains, each round of training overwrites weights that encoded prior knowledge. This is **catastrophic forgetting**. LoRA reduces this by training small adapter matrices, but each adapter merge dilutes prior adaptations: after N merges, each domain retains roughly $1/N$ of its contribution (the **O(1/N) dilution problem**). No existing method can identify *which specific weight groups* were damaged by new training, or selectively undo that damage.

2. Biological Inspiration

The Amazon Molly (*P. formosa*) reproduces asexually yet has thrived for ∼100,000 years, defying the prediction that without sexual recombination, harmful mutations accumulate irreversibly until extinction (Muller's Ratchet). A 2026 Nature paper revealed the mechanism: **gene conversion**—a repair process where the organism's two chromosome copies compare themselves, and when one carries a harmful mutation, the healthy copy overwrites it. MOLLI applies this principle to neural networks. The model's weights are the genome. Fine-tuning introduces "mutations." A quantized checkpoint serves as the healthy reference. And a Bayesian scoring system plays the role of natural selection, deciding which changes to keep and which to repair.

3. Method

Dual-strand encoding. We partition model parameters into L functional groups called **genes**—each attention head, MLP block, and LayerNorm is one gene ($L = 928$ for 7B parameters). Two copies are maintained: the **primary** $\mathbf{w}^p$ (live trainable weights) and the **complement** $\mathbf{w}^c$ (int16-quantized snapshot from before training). Overhead: 2 bytes/parameter.

Gene scoring. After fine-tuning, we estimate the effect of reverting each gene i on the loss $\mathcal{L}_j$ of prior domain j. The gradient of the loss tells us how sensitive each parameter is to small changes; the complement–primary difference tells us how much each parameter actually changed. Their dot product gives the estimated reversion effect:

$$\delta_{ij} \;=\; \left\langle \nabla_\theta \mathcal{L}_j,\;\; \mathbf{w}_i^c - \mathbf{w}_i^p \right\rangle \tag{1}$$

This is a first-order Taylor approximation. A single forward+backward pass yields $\nabla_\theta \mathcal{L}_j$ for the entire model, from which all $L = 928$ gene scores are extracted by slicing. Total cost: $\mathcal{O}(N_{\text{obj}})$ passes, not $\mathcal{O}(L \times N_{\text{obj}})$.

Why raw scores aren't enough. The score δ_{ij} depends on which evaluation samples we used—it is a noisy estimate. A gene might appear damaged purely by statistical fluctuation. Repairing based on noisy scores would introduce unnecessary changes. We need to separate genuine signal from noise.

Bayesian shrinkage. We apply **empirical Bayes** to obtain calibrated probabilities. We split evaluation data in half, score each half independently, and measure agreement (split-half correlation r). High agreement ($r \approx 1$) means the scores are reliable; low agreement means noise dominates. We decompose observed variance into signal τ^2 and noise σ^2, then compute a shrinkage factor:

$$\tau^2 = \max\!\big(0,\; \mathrm{Var}(\delta) - \sigma^2\big), \quad B = \frac{\tau^2}{\tau^2 + \sigma^2}, \quad \delta_{ij}^* = B\,\delta_{ij} + (1 - B)\,\mu_0 \tag{2}$$

When $B \approx 1$, data has high signal-to-noise and we trust observed scores. When $B \approx 0$, noise overwhelms signal and scores are shrunk toward the mean μ_0 (assuming the gene is unaffected). This prevents false-positive repairs. In our experiments, B ranged from 0.75 (SNR = 3.1) to 0.999 (SNR = 842.5), with split-half $r = 0.83$–0.999.

Repair decision. From the shrunk scores, we compute the posterior probability that gene i is deleterious to domain j:

$$P(\mathrm{del}_{ij}) = 1 - \Phi\left(\frac{-\delta_{ij}^*}{\sqrt{\mathrm{Var}_{\text{post}}}}\right), \quad \text{Repair if } \max_j P(\mathrm{del}_{ij}) - \alpha\, P(\mathrm{ben}_{i} \tag{3}$$

where Φ is the normal CDF, α balances prior-domain protection against current-domain benefit, and θ is the repair threshold. Repair copies $\mathbf{w}_i^c \to \mathbf{w}_i^p$, reverting that gene. Capped at 15% of genes per cycle. Genes with $P(\mathrm{del}) < 0.3$ on all priors are **fixed** ($\mathbf{w}_i^p \to \mathbf{w}_i^c$), permanently accepting the change.

4. Results

Qwen2.5-7B (928 genes), 100 train / 49 eval × 256 tokens, 3 epochs, bfloat16, 8-bit Adam.

Domains	MOLLI	LoRA	Δ	Time (M/L)
3 (code→legal→med)	**1.60**	2.10	−24%	38/15 m
5 (+ science, finance)	**1.11**	1.45	−23%	113/49 m
6 (+ general)	**1.52**	1.60	−5%	146/62 m

Table 1. GM perplexity (lower = better). Best: 1.11 at 5 domains.

	gen	code	leg	med	sci	fin	**GM**
MOLLI 5d	1.06	1.25	1.10	1.14	1.06	1.06	**1.11**
LoRA 5d	1.10	1.63	1.83	2.29	1.10	1.10	1.45
MOLLI 6d	1.05	1.53	2.53	2.80	1.05	1.05	**1.52**
LoRA 6d	1.11	1.68	2.81	2.65	1.11	1.11	1.60

Table 2. Per-domain PPL. At 5d, MOLLI retains all domains ≤ 1.25.

Key findings. (i) MOLLI achieves 23–24% lower GM PPL at 3 and 5 domains. (ii) At 5 domains, every prior domain stays ≤1.25; LoRA degrades to 2.29 (medical) and 1.83 (legal). (iii) At 6 domains the gap narrows to 5%, suggesting the fixed 15% repair cap becomes a bottleneck with 7 simultaneous objectives; an adaptive cap would likely restore the wider gap. (iv) 139/928 genes repaired per cycle (15% cap), 0–151 fixed—targeted, selective intervention.